ANO	Valor do ano	
MES	Valor do mês	
HEL	movimento de helicópteros	TATIC FLOW e BIMTRA (somente dados de SBBR)
RPC	Renda Per Capita	https://anuario.ipe.df.gov.br/home
POP	População	https://anuario.ipe.df.gov.br/home
FROTA	Frota	https://app.powerbi.com/view?r=eyJrIjoiMGQ2MjdiMWYtNzQxZS00Mzk2LWE0NmEtMzFiMTN
HELIPT	Helipontos	https://www.gov.br/anac/pt-br/assuntos/regulados/aeroportos-e-aerodromos/lista-de-aerodrom
QAV	Preço do QAV	https://www.gov.br/anp/pt-br/assuntos/precos-e-defesa-da-concorrencia/precos/precos-de-pro
QAVREF	Preço do QAV corrigido pelo IPCA para o mês de ref (mar/26)	
PAND	Dummy Pandemia	A partir de mar/20 = 1

JmNzI1OTk3IiwidCI6IjI1NzQ4ZjZlLWI0YTQtNGIyYi1hYjJhLWVmOTUyMjM2ODM2NiIsImMiOjR9

os-civis-cadastrados

dutores-e-importadores-de-derivados-de-petroleo-e-biodiesel

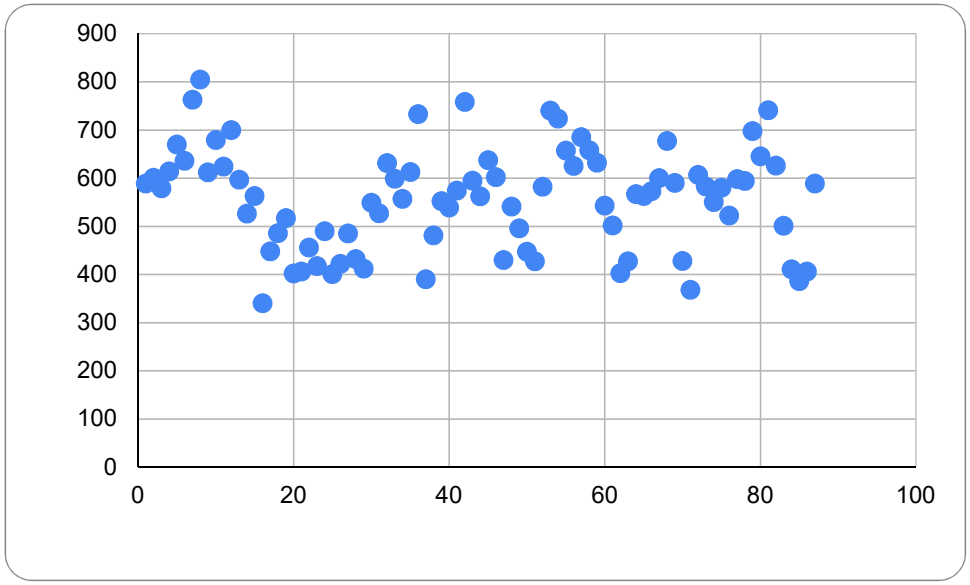
COD ICAO	MUNICIPIO	UF	TIPO
SBBR	BRASÍLIA	DF	AD PUB
SJXU	BRASÍLIA	DF	AD PRIV
SINA	BRASÍLIA	DF	AD PRIV
SSWB	BRASÍLIA	DF	AD PRIV
SDCB	BRASÍLIA	DF	AD PRIV
SSGP	BRASÍLIA	DF	AD PRIV
SWXU	BRASÍLIA	DF	AD PRIV
SIQE	BRASÍLIA	DF	AD PRIV
SWDF	BRASÍLIA	DF	HELIPT
SDHH	BRASÍLIA	DF	HELIPT
SDZA	BRASÍLIA	DF	HELIPT
SICA	BRASÍLIA	DF	HELIPT
SWSW	BRASÍLIA	DF	HELIPT
SDFG	BRASÍLIA	DF	HELIPT
SIIY	BRASÍLIA	DF	HELIPT
SNAW	BRASÍLIA	DF	HELIPT
SNCN	BRASÍLIA	DF	HELIPT
SSOH	BRASÍLIA	DF	HELIPT
SN2B	BRASÍLIA	DF	HELIPT
SI4J	BRASÍLIA	DF	HELIPT

ANO	MES	DATA	HEL	RPC	POP	FROTA	HELIPT	QAV	QAVREF	PAND
2019	1	1/1/2019	589	2047.86	3015000			6		0
2019	2	2/1/2019	601	2047.86	3015000			6		0
2019	3	3/1/2019	579	2047.86	3015000			6		0
2019	4	4/1/2019	614	2047.86	3015000			6		0
2019	5	5/1/2019	670	2047.86	3015000			6		0
2019	6	6/1/2019	636	2047.86	3015000			6		0
2019	7	7/1/2019	763	2047.86	3015000			6		0
2019	8	8/1/2019	805	2047.86	3015000			6		0
2019	9	9/1/2019	612	2047.86	3015000			6		0
2019	10	10/1/2019	679	2047.86	3015000			7		0
2019	11	11/1/2019	624	2047.86	3015000			7		0
2019	12	12/1/2019	700	2047.86	3015000			7		0
2020	1	1/1/2020	597		3052546			7		0
2020	2	2/1/2020	526		3052546			7		0
2020	3	3/1/2020	563		3052546			7		0
2020	4	4/1/2020	340		3052546			7		1
2020	5	5/1/2020	448		3052546			7		1
2020	6	6/1/2020	486		3052546			7		1
2020	7	7/1/2020	517		3052546			7		1
2020	8	8/1/2020	402		3052546			7		1
2020	9	9/1/2020	406		3052546			7		1
2020	10	10/1/2020	456		3052546			7		1

2020	11	11/1/2020	417		3052546	8			1
2020	12	12/1/2020	490		3052546	8			1
2021	1	1/1/2021	401	2320.55	3091667	8			1
2021	2	2/1/2021	422	2320.55	3091667	8			1
2021	3	3/1/2021	485	2320.55	3091667	8			1
2021	4	4/1/2021	432	2320.55	3091667	8			1
2021	5	5/1/2021	412	2320.55	3091667	8			1
2021	6	6/1/2021	549	2320.55	3091667	8			1
2021	7	7/1/2021	527	2320.55	3091667	8			1
2021	8	8/1/2021	631	2320.55	3091667	8			1
2021	9	9/1/2021	599	2320.55	3091667	8			1
2021	10	10/1/2021	557	2320.55	3091667	9			1
2021	11	11/1/2021	613	2320.55	3091667	9			1
2021	12	12/1/2021	733	2320.55	3091667	9			1
2022	1	1/1/2022	390		3130014	9			1
2022	2	2/1/2022	481		3130014	9			1
2022	3	3/1/2022	552		3130014	10	4.1585	4.968086	1
2022	4	4/1/2022	539		3130014	10	4.1585	4.915976	1
2022	5	5/1/2022	574		3130014	10			1
2022	6	6/1/2022	758		3130014	10			1
2022	7	7/1/2022	595		3130014	10			1
2022	8	8/1/2022	562		3130014	10			1
2022	9	9/1/2022	637		3130014	10			1

2022	10	10/1/2022	602		3130014	10			1
2022	11	11/1/2022	430		3130014	10			1
2022	12	12/1/2022	541		3130014	10			1
2023	1	1/1/2023	496		3167502	10			1
2023	2	2/1/2023	447		3167502	11			1
2023	3	3/1/2023	427		3167502	11			1
2023	4	4/1/2023	582		3167502	11			1
2023	5	5/1/2023	740		3167502	11			1
2023	6	6/1/2023	724		3167502	12			1
2023	7	7/1/2023	657		3167502	12	3.47054	3.927309	1
2023	8	8/1/2023	625		3167502	12	3.60729	4.072688	1
2023	9	9/1/2023	685		3167502	12	4.17923	4.706179	1
2023	10	10/1/2023	658		3167502	12	4.53222	5.091458	1
2023	11	11/1/2023	632		3167502	12	4.49149	5.031612	1
2023	12	12/1/2023	543		3167502	12	4.27567	4.763164	1
2024	1	1/1/2024	502	2579.39	3204070	12	3.82944	4.248213	1
2024	2	2/1/2024	403	2579.39	3204070	12	3.80717	4.188744	1
2024	3	3/1/2024	427	2579.39	3204070	12	4.05763	4.457176	1
2024	4	4/1/2024	567	2579.39	3204070	12	4.067	4.450559	1
2024	5	5/1/2024	563	2579.39	3204070	12	4.14394	4.513989	1
2024	6	6/1/2024	573	2579.39	3204070	12	3.92496	4.266493	1
2024	7	7/1/2024	600	2579.39	3204070	12	3.98482	4.315163	1
2024	8	8/1/2024	677	2579.39	3204070	12	4.15077	4.495767	1

2024	9	9/1/2024	590	2579.39	3204070			12	4.02015	4.335217	1
2024	10	10/1/2024	428	2579.39	3204070			12	3.71561	3.984499	1
2024	11	11/1/2024	368	2579.39	3204070			12	3.66252	3.91231	1
2024	12	12/1/2024	607	2579.39	3204070			12	3.76972	4.00599	1
2025	1	1/1/2025	583					12	3.94139	4.18173	1
2025	2	2/1/2025	550					12	4.32445	4.528818	1
2025	3	3/1/2025	580					12	4.08082	4.249874	1
2025	4	4/1/2025	522					12	3.75203	3.89073	1
2025	5	5/1/2025	598					12	3.73178	3.859696	1
2025	6	6/1/2025	594					12	3.51021	3.621838	1
2025	7	7/1/2025	698					12	3.53514	3.638104	1
2025	8	8/1/2025	645					12	3.67885	3.790171	1
2025	9	9/1/2025	741					12	3.57641	3.667032	1
2025	10	10/1/2025	626					12	3.58344	3.670938	1
2025	11	11/1/2025	501					12	3.62173	3.703496	1
2025	12	12/1/2025	411					12	3.76103	3.833291	1
2026	1	1/1/2026	386				492	12	3.47703	3.532179	1
2026	2	2/1/2026	406				492	12	3.41284	3.442872	1
2026	3	3/1/2026	589				492	12	3.65865	3.65865	1


900
800
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0
20
40
60
80
100

	DATA	*HEL*	*RPC*	*POP*	*FROTA*	*HELIPT*	*QAV*	*PAND*
DATA	1							
HEL	-0.04054	1						
RPC	0.982008	-0.50421	1					
POP	0.986051	-0.10708	0.992354	1				
FROTA	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	1			
HELIPT	0.960956	0.010908	0.971564	0.973444	#DIV/0!	1		
QAV	-0.57435	0.186621	#DIV/0!	-0.32363	#DIV/0!	-0.23595	1	
PAND	0.65438	-0.34737	0.873449	0.693346	#DIV/0!	0.660932	#DIV/0!	1

ANO	MES	DATA	HEL
2019	1	1/1/2019	589
2019	2	2/1/2019	601
2019	3	3/1/2019	579
2019	4	4/1/2019	614
2019	5	5/1/2019	670
2019	6	6/1/2019	636
2019	7	7/1/2019	763
2019	8	8/1/2019	805
2019	9	9/1/2019	612
2019	10	10/1/2019	679
2019	11	11/1/2019	624
2019	12	12/1/2019	700
2020	1	1/1/2020	597
2020	2	2/1/2020	526
2020	3	3/1/2020	563
2020	4	4/1/2020	340
2020	5	5/1/2020	448
2020	6	6/1/2020	486
2020	7	7/1/2020	517
2020	8	8/1/2020	402
2020	9	9/1/2020	406
2020	10	10/1/2020	456

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.040536515
R Square	0.001643209
Adjusted R Squ	-0.010102165
Standard Error	106.0076878
Observations	87

ANOVA

	df	*SS*	*MS*	*F*
Regression	1	1572.174292	1572.174292	0.139902658
Residual	85	955198.5384	11237.62986	
Total	86	956770.7126		

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	806.3684448	665.7492031	1.211219542	0.229168015
DATA	-0.005560731	0.014866848	-0.374035638	0.709309207

RESIDUAL OUTPUT

2020	11	11/1/2020	417
2020	12	12/1/2020	490
2021	1	1/1/2021	401
2021	2	2/1/2021	422
2021	3	3/1/2021	485
2021	4	4/1/2021	432
2021	5	5/1/2021	412
2021	6	6/1/2021	549
2021	7	7/1/2021	527
2021	8	8/1/2021	631
2021	9	9/1/2021	599
2021	10	10/1/2021	557
2021	11	11/1/2021	613
2021	12	12/1/2021	733
2022	1	1/1/2022	390
2022	2	2/1/2022	481
2022	3	3/1/2022	552
2022	4	4/1/2022	539
2022	5	5/1/2022	574
2022	6	6/1/2022	758
2022	7	7/1/2022	595
2022	8	8/1/2022	562
2022	9	9/1/2022	637

Observation	*Predicted HEL*	*Residuals*	*andard Residuals*
1	564.6657108	24.33428917	0.230898476
2	564.4933282	36.50667183	0.346397415
3	564.3376277	14.6623723	0.13912547
4	564.165245	49.83475496	0.472862341
5	563.9984231	106.0015769	1.005807169
6	563.8260405	72.17395955	0.684830246
7	563.6592185	199.3407815	1.89146608
8	563.4868359	241.5131641	2.291623192
9	563.3144532	48.6855468	0.461957958
10	563.1476313	115.8523687	1.099277449
11	562.9752486	61.02475139	0.579039805
12	562.8084267	137.1915733	1.301756748
13	562.636044	34.36395598	0.326066029
14	562.4636614	-36.46366135	-0.345989306
15	562.3024002	0.697599845	0.00661925
16	562.1300175	-222.1300175	-2.107704156
17	561.9631956	-113.9631956	-1.08135183
18	561.7908129	-75.7908129	-0.719149141
19	561.623991	-44.62399097	-0.423419456
20	561.4516083	-159.4516083	-1.512973444
21	561.2792256	-155.2792256	-1.473383349

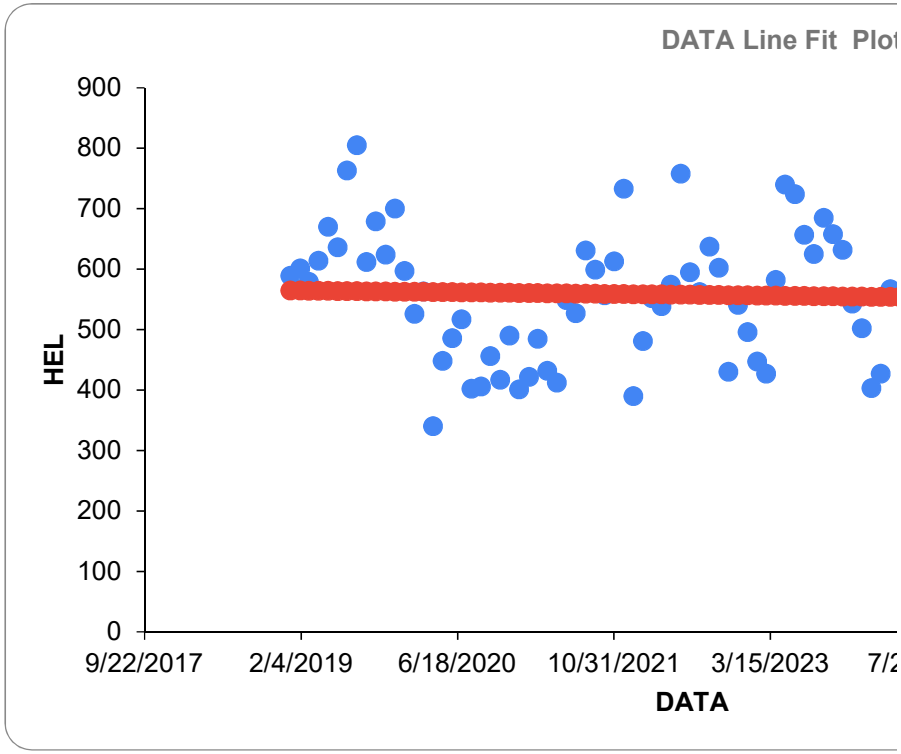
2022	10	10/1/2022	602
2022	11	11/1/2022	430
2022	12	12/1/2022	541
2023	1	1/1/2023	496
2023	2	2/1/2023	447
2023	3	3/1/2023	427
2023	4	4/1/2023	582
2023	5	5/1/2023	740
2023	6	6/1/2023	724
2023	7	7/1/2023	657
2023	8	8/1/2023	625
2023	9	9/1/2023	685
2023	10	10/1/2023	658
2023	11	11/1/2023	632
2023	12	12/1/2023	543
2024	1	1/1/2024	502
2024	2	2/1/2024	403
2024	3	3/1/2024	427
2024	4	4/1/2024	567
2024	5	5/1/2024	563
2024	6	6/1/2024	573
2024	7	7/1/2024	600
2024	8	8/1/2024	677

22	561.1124037	-105.1124037	-0.997370156
23	560.9400211	-143.9400211	-1.365790108
24	560.7731991	-70.77319913	-0.671538982
25	560.6008165	-159.6008165	-1.514389221
26	560.4284338	-138.4284338	-1.31349283
27	560.2727333	-75.27273334	-0.714233288
28	560.1003507	-128.1003507	-1.215493721
29	559.9335287	-147.9335287	-1.403682928
30	559.7611461	-10.76114608	-0.102108272
31	559.5943242	-32.59432415	-0.309274691
32	559.4219415	71.57805851	0.679175976
33	559.2495588	39.75044117	0.377176264
34	559.0827369	-2.082736902	-0.019762269
35	558.9103542	54.08964576	0.513235322
36	558.7435323	174.2564677	1.653450917
37	558.5711496	-168.5711496	-1.599505176
38	558.398767	-77.39876699	-0.734406384
39	558.2430665	-6.24306652	-0.059237997
40	558.0706839	-19.07068386	-0.1809542
41	557.9038619	16.09613807	0.152729908
42	557.7314793	200.2685207	1.900269032
43	557.5646573	37.43534266	0.355209207
44	557.3922747	4.607725324	0.043720889

2024	9	9/1/2024	590
2024	10	10/1/2024	428
2024	11	11/1/2024	368
2024	12	12/1/2024	607
2025	1	1/1/2025	583
2025	2	2/1/2025	550
2025	3	3/1/2025	580
2025	4	4/1/2025	522
2025	5	5/1/2025	598
2025	6	6/1/2025	594
2025	7	7/1/2025	698
2025	8	8/1/2025	645
2025	9	9/1/2025	741
2025	10	10/1/2025	626
2025	11	11/1/2025	501
2025	12	12/1/2025	411
2026	1	1/1/2026	386
2026	2	2/1/2026	406
2026	3	3/1/2026	589

45	557.219892	79.78010799	0.757001989
46	557.0530701	44.94692992	0.426483697
47	556.8806874	-126.8806874	-1.203920817
48	556.7138655	-15.71386549	-0.149102674
49	556.5414828	-60.54148283	-0.574454261
50	556.3691002	-109.3691002	-1.03776027
51	556.2133997	-129.2133997	-1.226055004
52	556.041017	25.95898296	0.246314554
53	555.8741951	184.1258049	1.747097166
54	555.7018124	168.2981876	1.596915146
55	555.5349905	101.4650095	0.96276147
56	555.3626079	69.63739214	0.660761758
57	555.1902252	129.8097748	1.231713772
58	555.0234033	102.9765967	0.977104325
59	554.8510206	77.14897939	0.732036248
60	554.6841987	-11.68419868	-0.110866754
61	554.511816	-52.51181601	-0.498263918
62	554.3394334	-151.3394334	-1.436000214
63	554.1781722	-127.1781722	-1.206743532
64	554.0057895	12.99421051	0.12329694
65	553.8389676	9.161032437	0.086925425
66	553.6665849	19.3334151	0.183447153
67	553.499763	46.50023703	0.441222415
68	553.3273803	123.6726197	1.173480727

69	553.1549976	36.84500235	0.3496077
70	552.9881757	-124.9881757	-1.18596352
71	552.8157931	-184.8157931	-1.753644192
72	552.6489711	54.35102887	0.515715484
73	552.4765885	30.52341153	0.289624617
74	552.3042058	-2.304205805	-0.0218637
75	552.1485053	27.85149466	0.264271852
76	551.9761227	-29.97612268	-0.284431609
77	551.8093007	46.19069925	0.438285333
78	551.6369181	42.36308192	0.401966582
79	551.4700962	146.5299038	1.390364485
80	551.2977135	93.70228651	0.889104052
81	551.1253308	189.8746692	1.801645873
82	550.9585089	75.0414911	0.712039122
83	550.7861262	-49.78612624	-0.472400922
84	550.6193043	-139.6193043	-1.32479253
85	550.4469216	-164.4469216	-1.560372002
86	550.274539	-144.274539	-1.368964217
87	550.1188385	38.88116148	0.368928011



Significance F
0.709309207

Lower 95%	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
-517.3193303	2130.05622	-517.3193303	2130.05622
-0.035120008	0.023998546	-0.035120008	0.023998546

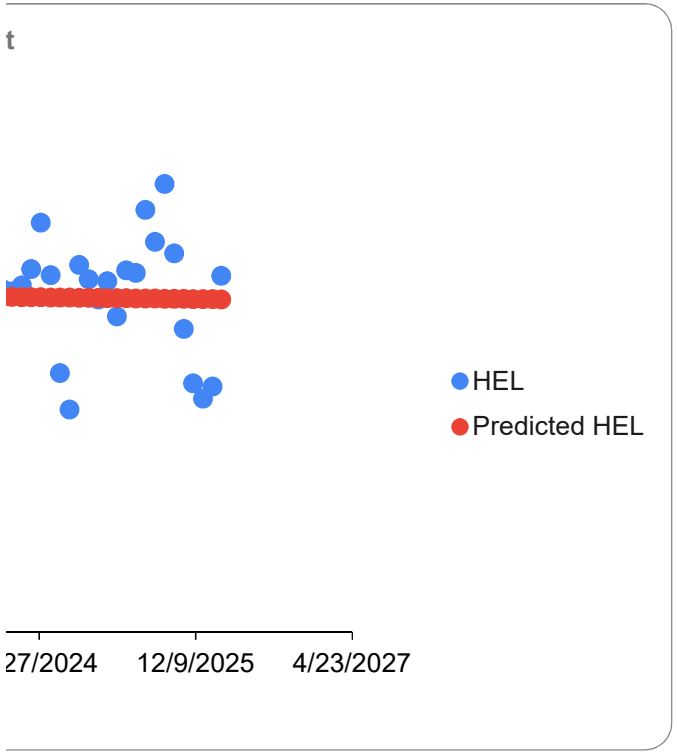
PROBABILITY OUTPUT

Percentile	*HEL*
0.574712644	340
1.724137931	368
2.873563218	386
4.022988506	390
5.172413793	401
6.32183908	402
7.471264368	403
8.620689655	406
9.770114943	406
10.91954023	411
12.06896552	412
13.2183908	417
14.36781609	422
15.51724138	427
16.66666667	427
17.81609195	428
18.96551724	430
20.11494253	432
21.26436782	447
22.4137931	448
23.56321839	456

24.71264368	481
25.86206897	485
27.01149425	486
28.16091954	490
29.31034483	496
30.45977011	501
31.6091954	502
32.75862069	517
33.90804598	522
35.05747126	526
36.20689655	527
37.35632184	539
38.50574713	541
39.65517241	543
40.8045977	549
41.95402299	550
43.10344828	552
44.25287356	557
45.40229885	562
46.55172414	563
47.70114943	563
48.85057471	567
50	573

51.14942529	574
52.29885057	579
53.44827586	580
54.59770115	582
55.74712644	583
56.89655172	589
58.04597701	589
59.1954023	590
60.34482759	594
61.49425287	595
62.64367816	597
63.79310345	598
64.94252874	599
66.09195402	600
67.24137931	601
68.3908046	602
69.54022989	607
70.68965517	612
71.83908046	613
72.98850575	614
74.13793103	624
75.28735632	625
76.43678161	626
77.5862069	631

78.73563218	632
79.88505747	636
81.03448276	637
82.18390805	645
83.33333333	657
84.48275862	658
85.63218391	670
86.7816092	677
87.93103448	679
89.08045977	685
90.22988506	698
91.37931034	700
92.52873563	724
93.67816092	733
94.82758621	740
95.97701149	741
97.12643678	758
98.27586207	763
99.42528736	805


t
HEL
Predicted HEL
27/2024
12/9/2025
4/23/2027

ANO	MES	DATA	HEL	DM1	DM2	DM3	DM4	DM5	DM6
2019	1	1/1/2019	589	1	0	0	0	0	0
2019	2	2/1/2019	601	0	1	0	0	0	0
2019	3	3/1/2019	579	0	0	1	0	0	0
2019	4	4/1/2019	614	0	0	0	1	0	0
2019	5	5/1/2019	670	0	0	0	0	1	0
2019	6	6/1/2019	636	0	0	0	0	0	1
2019	7	7/1/2019	763	0	0	0	0	0	0
2019	8	8/1/2019	805	0	0	0	0	0	0
2019	9	9/1/2019	612	0	0	0	0	0	0
2019	10	10/1/2019	679	0	0	0	0	0	0
2019	11	11/1/2019	624	0	0	0	0	0	0
2019	12	12/1/2019	700	0	0	0	0	0	0
2020	1	1/1/2020	597	1	0	0	0	0	0
2020	2	2/1/2020	526	0	1	0	0	0	0
2020	3	3/1/2020	563	0	0	1	0	0	0
2020	4	4/1/2020	340	0	0	0	1	0	0
2020	5	5/1/2020	448	0	0	0	0	1	0
2020	6	6/1/2020	486	0	0	0	0	0	1
2020	7	7/1/2020	517	0	0	0	0	0	0
2020	8	8/1/2020	402	0	0	0	0	0	0
2020	9	9/1/2020	406	0	0	0	0	0	0
2020	10	10/1/2020	456	0	0	0	0	0	0

2020	11	11/1/2020	417	0	0	0	0	0	0
2020	12	12/1/2020	490	0	0	0	0	0	0
2021	1	1/1/2021	401	1	0	0	0	0	0
2021	2	2/1/2021	422	0	1	0	0	0	0
2021	3	3/1/2021	485	0	0	1	0	0	0
2021	4	4/1/2021	432	0	0	0	1	0	0
2021	5	5/1/2021	412	0	0	0	0	1	0
2021	6	6/1/2021	549	0	0	0	0	0	1
2021	7	7/1/2021	527	0	0	0	0	0	0
2021	8	8/1/2021	631	0	0	0	0	0	0
2021	9	9/1/2021	599	0	0	0	0	0	0
2021	10	10/1/2021	557	0	0	0	0	0	0
2021	11	11/1/2021	613	0	0	0	0	0	0
2021	12	12/1/2021	733	0	0	0	0	0	0
2022	1	1/1/2022	390	1	0	0	0	0	0
2022	2	2/1/2022	481	0	1	0	0	0	0
2022	3	3/1/2022	552	0	0	1	0	0	0
2022	4	4/1/2022	539	0	0	0	1	0	0
2022	5	5/1/2022	574	0	0	0	0	1	0
2022	6	6/1/2022	758	0	0	0	0	0	1
2022	7	7/1/2022	595	0	0	0	0	0	0
2022	8	8/1/2022	562	0	0	0	0	0	0
2022	9	9/1/2022	637	0	0	0	0	0	0

2022	10	10/1/2022	602	0	0	0	0	0	0
2022	11	11/1/2022	430	0	0	0	0	0	0
2022	12	12/1/2022	541	0	0	0	0	0	0
2023	1	1/1/2023	496	1	0	0	0	0	0
2023	2	2/1/2023	447	0	1	0	0	0	0
2023	3	3/1/2023	427	0	0	1	0	0	0
2023	4	4/1/2023	582	0	0	0	1	0	0
2023	5	5/1/2023	740	0	0	0	0	1	0
2023	6	6/1/2023	724	0	0	0	0	0	1
2023	7	7/1/2023	657	0	0	0	0	0	0
2023	8	8/1/2023	625	0	0	0	0	0	0
2023	9	9/1/2023	685	0	0	0	0	0	0
2023	10	10/1/2023	658	0	0	0	0	0	0
2023	11	11/1/2023	632	0	0	0	0	0	0
2023	12	12/1/2023	543	0	0	0	0	0	0
2024	1	1/1/2024	502	1	0	0	0	0	0
2024	2	2/1/2024	403	0	1	0	0	0	0
2024	3	3/1/2024	427	0	0	1	0	0	0
2024	4	4/1/2024	567	0	0	0	1	0	0
2024	5	5/1/2024	563	0	0	0	0	1	0
2024	6	6/1/2024	573	0	0	0	0	0	1
2024	7	7/1/2024	600	0	0	0	0	0	0
2024	8	8/1/2024	677	0	0	0	0	0	0

2024	9	9/1/2024	590	0	0	0	0	0	0
2024	10	10/1/2024	428	0	0	0	0	0	0
2024	11	11/1/2024	368	0	0	0	0	0	0
2024	12	12/1/2024	607	0	0	0	0	0	0
2025	1	1/1/2025	583	1	0	0	0	0	0
2025	2	2/1/2025	550	0	1	0	0	0	0
2025	3	3/1/2025	580	0	0	1	0	0	0
2025	4	4/1/2025	522	0	0	0	1	0	0
2025	5	5/1/2025	598	0	0	0	0	1	0
2025	6	6/1/2025	594	0	0	0	0	0	1
2025	7	7/1/2025	698	0	0	0	0	0	0
2025	8	8/1/2025	645	0	0	0	0	0	0
2025	9	9/1/2025	741	0	0	0	0	0	0
2025	10	10/1/2025	626	0	0	0	0	0	0
2025	11	11/1/2025	501	0	0	0	0	0	0
2025	12	12/1/2025	411	0	0	0	0	0	0
2026	1	1/1/2026	386	1	0	0	0	0	0
2026	2	2/1/2026	406	0	1	0	0	0	0
2026	3	3/1/2026	589	0	0	1	0	0	0
				1	2	3	4	5	6

DM7	DM8	DM9	DM10	DM11	DM12
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.48515479
R Square	0.235375171
Adjusted R Squ	0.109896862
Standard Error	98.76373443
Observations	87

ANOVA

	df	*SS*
Regression	12	225200.0698
Residual	75	731570.6429
Total	87	956770.7126

	Coefficients	*Standard Error*
Intercept	513.7142857	37.32918284
DM1	-20.71428571	51.11508873
DM2	-34.21428571	51.11508873
DM3	11.53571429	51.11508873
DM4	0	0
DM5	58.42857143	52.79143664

0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0

DM6	103.4285714	52.79143664
DM7	108.7142857	52.79143664
DM8	107.2857143	52.79143664
DM9	96.28571429	52.79143664
DM10	58.57142857	52.79143664
DM11	-1.571428571	52.79143664
DM12	61.28571429	52.79143664

RESIDUAL OUTPUT

Observation	*Predicted HEL*	*Residuals*
1	493	96
2	479.5	121.5
3	525.25	53.75
4	513.7142857	100.2857143
5	572.1428571	97.85714286
6	617.1428571	18.85714286
7	622.4285714	140.5714286
8	621	184
9	610	2
10	572.2857143	106.7142857

0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0

11	512.1428571	111.8571429
12	575	125
13	493	104
14	479.5	46.5
15	525.25	37.75
16	513.7142857	-173.7142857
17	572.1428571	-124.1428571
18	617.1428571	-131.1428571
19	622.4285714	-105.4285714
20	621	-219
21	610	-204
22	572.2857143	-116.2857143
23	512.1428571	-95.14285714
24	575	-85
25	493	-92
26	479.5	-57.5
27	525.25	-40.25
28	513.7142857	-81.71428571
29	572.1428571	-160.1428571
30	617.1428571	-68.14285714
31	622.4285714	-95.42857143
32	621	10
33	610	-11

0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
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0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
7	8	9	10	11	12

34	572.2857143	-15.28571429
35	512.1428571	100.8571429
36	575	158
37	493	-103
38	479.5	1.5
39	525.25	26.75
40	513.7142857	25.28571429
41	572.1428571	1.857142857
42	617.1428571	140.8571429
43	622.4285714	-27.42857143
44	621	-59
45	610	27
46	572.2857143	29.71428571
47	512.1428571	-82.14285714
48	575	-34
49	493	3
50	479.5	-32.5
51	525.25	-98.25
52	513.7142857	68.28571429
53	572.1428571	167.8571429
54	617.1428571	106.8571429
55	622.4285714	34.57142857
56	621	4
57	610	75

58	572.2857143	85.71428571
59	512.1428571	119.8571429
60	575	-32
61	493	9
62	479.5	-76.5
63	525.25	-98.25
64	513.7142857	53.28571429
65	572.1428571	-9.142857143
66	617.1428571	-44.14285714
67	622.4285714	-22.42857143
68	621	56
69	610	-20
70	572.2857143	-144.2857143
71	512.1428571	-144.1428571
72	575	32
73	493	90
74	479.5	70.5
75	525.25	54.75
76	513.7142857	8.285714286
77	572.1428571	25.85714286
78	617.1428571	-23.14285714
79	622.4285714	75.57142857
80	621	24
81	610	131
82	572.2857143	53.71428571
83	512.1428571	-11.14285714
84	575	-164
85	493	-107
86	479.5	-73.5
87	525.25	63.75

MS	*F*	*Significance F*
18766.67248	2.098847235	0.02701813
9754.275238		

t Stat	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
13.76173403	3.26661E-22	439.3507402	588.0778312	439.3507402	588.0778312
-0.405247965	0.686448906	-122.5407641	81.11219265	-122.5407641	81.11219265
-0.669357846	0.505322631	-136.0407641	67.61219265	-136.0407641	67.61219265
0.225681195	0.82206324	-90.29076408	113.3621926	-90.29076408	113.3621926
65535	#NUM!	0	0	0	0
1.106781235	#NUM!	-46.73736321	163.5945061	-46.73736321	163.5945061

1.95919221	0.053807237	-1.737363211	208.5945061	-1.737363211	208.5945061
2.059316674	0.042935418	3.548351075	213.8802204	3.548351075	213.8802204
2.032256008	0.045668907	2.119779646	212.4516489	2.119779646	212.4516489
1.823888881	0.072152506	-8.880220354	201.4516489	-8.880220354	201.4516489
1.109487301	0.270764858	-46.59450607	163.7373632	-46.59450607	163.7373632
-0.029766732	0.976332159	-106.7373632	103.5945061	-106.7373632	103.5945061
1.160902567	0.249363243	-43.88022035	166.4516489	-43.88022035	166.4516489

PROBABILITY OUTPUT

andard Residuals	*Percentile*	*HEL*
1.046894033	0.574712644	340
1.324975261	1.724137931	368
0.586151607	2.873563218	386
1.093630374	4.022988506	390
1.067146448	5.172413793	401
0.205639899	6.32183908	402
1.532951978	7.471264368	403
2.006546898	8.620689655	406
0.021810292	9.770114943	406
1.163734885	10.91954023	411

1.219818494	12.06896552	412
1.363143273	13.2183908	417
1.134135203	14.36781609	422
0.507089297	15.51724138	427
0.411669268	16.66666667	427
-1.89437968	17.81609195	428
-1.353796005	18.96551724	430
-1.430132028	20.11494253	432
-1.149713983	21.26436782	447
-2.388227014	22.4137931	448
-2.224649821	23.56321839	456
-1.268112713	24.71264368	481
-1.037546765	25.86206897	485
-0.926937425	27.01149425	486
-1.003273449	28.16091954	490
-0.627045905	29.31034483	496
-0.438932134	30.45977011	501
-0.891106231	31.6091954	502
-1.746381267	32.75862069	517
-0.743107818	33.90804598	522
-1.040662521	35.05747126	526
0.109051462	36.20689655	527
-0.119956608	37.35632184	539

-0.166692949	38.50574713	541
1.099861886	39.65517241	543
1.723013097	40.8045977	549
-1.123230057	41.95402299	550
0.016357719	43.10344828	552
0.29171266	44.25287356	557
0.275744411	45.40229885	562
0.020252414	46.55172414	563
1.536067734	47.70114943	563
-0.299112581	48.85057471	567
-0.643403625	50	573
0.294438947	51.14942529	574
0.324038629	52.29885057	579
-0.895779865	53.44827586	580
-0.37077497	54.59770115	582
0.032715439	55.74712644	583
-0.354417251	56.89655172	589
-1.071430612	58.04597701	589
0.744665696	59.1954023	590
1.830506681	60.34482759	594
1.165292763	61.49425287	595
0.377006482	62.64367816	597
0.043620585	63.79310345	598
0.817885964	64.94252874	599

0.934726816	66.09195402	600
1.307059664	67.24137931	601
-0.348964678	68.3908046	602
0.098146316	69.54022989	607
-0.834243683	70.68965517	612
-1.071430612	71.83908046	613
0.581088504	72.98850575	614
-0.099704194	74.13793103	624
-0.48138431	75.28735632	625
-0.24458685	76.43678161	626
0.610688186	77.5862069	631
-0.218102924	78.73563218	632
-1.573456806	79.88505747	636
-1.571898928	81.03448276	637
0.348964678	82.18390805	645
0.981463156	83.33333333	657
0.768812806	84.48275862	658
0.597056753	85.63218391	670
0.090356926	86.7816092	677
0.281975923	87.93103448	679
-0.25237624	89.08045977	685
0.824117476	90.22988506	698
0.261723508	91.37931034	700
1.42857415	92.52873563	724
0.585762138	93.67816092	733
-0.121514486	94.82758621	740
-1.788443974	95.97701149	741
-1.166850641	97.12643678	758
-0.801528244	98.27586207	763
0.695203069	99.42528736	805

ANO	MES	DATA	HEL	POP
2019	1	1/1/2019	589	3015000
2019	2	2/1/2019	601	3015000
2019	3	3/1/2019	579	3015000
2019	4	4/1/2019	614	3015000
2019	5	5/1/2019	670	3015000
2019	6	6/1/2019	636	3015000
2019	7	7/1/2019	763	3015000
2019	8	8/1/2019	805	3015000
2019	9	9/1/2019	612	3015000
2019	10	10/1/2019	679	3015000
2019	11	11/1/2019	624	3015000
2019	12	12/1/2019	700	3015000
2020	1	1/1/2020	597	3052546
2020	2	2/1/2020	526	3052546
2020	3	3/1/2020	563	3052546
2020	4	4/1/2020	340	3052546
2020	5	5/1/2020	448	3052546
2020	6	6/1/2020	486	3052546
2020	7	7/1/2020	517	3052546
2020	8	8/1/2020	402	3052546
2020	9	9/1/2020	406	3052546
2020	10	10/1/2020	456	3052546

2020	11	11/1/2020	417	3052546
2020	12	12/1/2020	490	3052546
2021	1	1/1/2021	401	3091667
2021	2	2/1/2021	422	3091667
2021	3	3/1/2021	485	3091667
2021	4	4/1/2021	432	3091667
2021	5	5/1/2021	412	3091667
2021	6	6/1/2021	549	3091667
2021	7	7/1/2021	527	3091667
2021	8	8/1/2021	631	3091667
2021	9	9/1/2021	599	3091667
2021	10	10/1/2021	557	3091667
2021	11	11/1/2021	613	3091667
2021	12	12/1/2021	733	3091667
2022	1	1/1/2022	390	3130014
2022	2	2/1/2022	481	3130014
2022	3	3/1/2022	552	3130014
2022	4	4/1/2022	539	3130014
2022	5	5/1/2022	574	3130014
2022	6	6/1/2022	758	3130014
2022	7	7/1/2022	595	3130014
2022	8	8/1/2022	562	3130014
2022	9	9/1/2022	637	3130014

2022	10	10/1/2022	602	3130014
2022	11	11/1/2022	430	3130014
2022	12	12/1/2022	541	3130014
2023	1	1/1/2023	496	3167502
2023	2	2/1/2023	447	3167502
2023	3	3/1/2023	427	3167502
2023	4	4/1/2023	582	3167502
2023	5	5/1/2023	740	3167502
2023	6	6/1/2023	724	3167502
2023	7	7/1/2023	657	3167502
2023	8	8/1/2023	625	3167502
2023	9	9/1/2023	685	3167502
2023	10	10/1/2023	658	3167502
2023	11	11/1/2023	632	3167502
2023	12	12/1/2023	543	3167502
2024	1	1/1/2024	502	3204070
2024	2	2/1/2024	403	3204070
2024	3	3/1/2024	427	3204070
2024	4	4/1/2024	567	3204070
2024	5	5/1/2024	563	3204070
2024	6	6/1/2024	573	3204070
2024	7	7/1/2024	600	3204070
2024	8	8/1/2024	677	3204070

2024	9	9/1/2024	590	3204070
2024	10	10/1/2024	428	3204070
2024	11	11/1/2024	368	3204070
2024	12	12/1/2024	607	3204070

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.107080609
R Square	0.011466257
Adjusted R Squ	-0.002655654
Standard Error	106.7850294
Observations	72

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	9258.677301	9258.677301	0.811947978	0.370634793
Residual	70	798212.9755	11403.04251		
Total	71	807471.6528			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*
Intercept	1100.439625	603.8601506	1.822341852	0.072674437	-103.9214697	2304.80072	-103.9214697
POP	-0.000174915	0.000194117	-0.90108156	0.370634793	-0.000562069	0.000212239	-0.000562069

RESIDUAL OUTPUT

PROBABILITY OUTPUT

Observation	Predicted HEL	Residuals	andard Residuals
1	573.0707773	15.92922273	0.150232664
2	573.0707773	27.92922273	0.263407801
3	573.0707773	5.929222727	0.05592005
4	573.0707773	40.92922273	0.386014199
5	573.0707773	96.92922273	0.914164838
6	573.0707773	62.92922273	0.59350195
7	573.0707773	189.9292227	1.79127215
8	573.0707773	231.9292227	2.187385129
9	573.0707773	38.92922273	0.367151676
10	573.0707773	105.9292227	0.999046191
11	573.0707773	50.92922273	0.480326813
12	573.0707773	126.9292227	1.197102681
13	566.5034172	30.49658284	0.287621245
14	566.5034172	-40.50341716	-0.381998315
15	566.5034172	-3.503417155	-0.033041643
16	566.5034172	-226.5034172	-2.136212938
17	566.5034172	-118.5034172	-1.117636706
18	566.5034172	-80.50341716	-0.759248772
19	566.5034172	-49.50341716	-0.466879668
20	566.5034172	-164.5034172	-1.551474731
21	566.5034172	-160.5034172	-1.513749685

Percentile	HEL
0.694444444	340
2.083333333	368
3.472222222	390
4.861111111	401
6.25	402
7.638888889	403
9.027777778	406
10.41666667	412
11.80555556	417
13.19444444	422
14.58333333	427
15.97222222	427
17.36111111	428
18.75	430
20.13888889	432
21.52777778	447
22.91666667	448
24.30555556	456
25.69444444	481
27.08333333	485
28.47222222	486

22	566.5034172	-110.5034172	-1.042186614	29.86111111	490
23	566.5034172	-149.5034172	-1.41000581	31.25	496
24	566.5034172	-76.50341716	-0.721523726	32.63888889	502
25	559.6605658	-158.6605658	-1.496369273	34.02777778	517
26	559.6605658	-137.6605658	-1.298312783	35.41666667	526
27	559.6605658	-74.66056585	-0.704143314	36.80555556	527
28	559.6605658	-127.6605658	-1.204000169	38.19444444	539
29	559.6605658	-147.6605658	-1.392625397	39.58333333	541
30	559.6605658	-10.66056585	-0.100542583	40.97222222	543
31	559.6605658	-32.66056585	-0.308030334	42.36111111	549
32	559.6605658	71.33943415	0.672820853	43.75	552
33	559.6605658	39.33943415	0.371020487	45.13888889	557
34	559.6605658	-2.660565849	-0.025092492	46.52777778	562
35	559.6605658	53.33943415	0.503058147	47.91666667	563
36	559.6605658	173.3394342	1.634809517	49.30555556	563
37	552.9530988	-162.9530988	-1.536853273	50.69444444	567
38	552.9530988	-71.95309878	-0.678608484	52.08333333	573
39	552.9530988	-0.953098784	-0.008988924	53.47222222	574
40	552.9530988	-13.95309878	-0.131595322	54.86111111	579
41	552.9530988	21.04690122	0.198498827	56.25	582
42	552.9530988	205.0469012	1.933850928	57.63888889	589
43	552.9530988	42.04690122	0.396555317	59.02777778	590
44	552.9530988	9.046901216	0.08532369	60.41666667	595

45	552.9530988	84.04690122	0.792668296	61.80555556	597
46	552.9530988	49.04690122	0.462574147	63.19444444	599
47	552.9530988	-122.9530988	-1.159602816	64.58333333	600
48	552.9530988	-11.95309878	-0.112732799	65.97222222	601
49	546.3958837	-50.39588374	-0.475296754	67.36111111	602
50	546.3958837	-99.39588374	-0.937428563	68.75	607
51	546.3958837	-119.3958837	-1.126053791	70.13888889	612
52	546.3958837	35.60411626	0.335791728	71.52777778	613
53	546.3958837	193.6041163	1.825931031	72.91666667	614
54	546.3958837	177.6041163	1.675030849	74.30555556	624
55	546.3958837	110.6041163	1.043136334	75.69444444	625
56	546.3958837	78.60411626	0.741335969	77.08333333	631
57	546.3958837	138.6041163	1.307211654	78.47222222	632
58	546.3958837	111.6041163	1.052567595	79.86111111	636
59	546.3958837	85.60411626	0.807354799	81.25	637
60	546.3958837	-3.395883739	-0.032027467	82.63888889	657
61	539.9995905	-37.99959053	-0.358384072	84.02777778	658
62	539.9995905	-136.9995905	-1.292078952	85.41666667	670
63	539.9995905	-112.9995905	-1.065728678	86.80555556	677
64	539.9995905	27.00040947	0.25464792	88.19444444	679
65	539.9995905	23.00040947	0.216922874	89.58333333	685
66	539.9995905	33.00040947	0.311235488	90.97222222	700
67	539.9995905	60.00040947	0.565879547	92.36111111	724
68	539.9995905	137.0004095	1.292086676	93.75	733
69	539.9995905	50.00040947	0.471566933	95.13888889	740
70	539.9995905	-111.9995905	-1.056297417	96.52777778	758
71	539.9995905	-171.9995905	-1.622173101	97.91666667	763
72	539.9995905	67.00040947	0.631898377	99.30555556	805

Upper 95,0%
2304.80072
0.000212239

ANO	MES	DATA	HEL	HELIPT	PAND
2019	1	1/1/2019	589	6	0
2019	2	2/1/2019	601	6	0
2019	3	3/1/2019	579	6	0
2019	4	4/1/2019	614	6	0
2019	5	5/1/2019	670	6	0
2019	6	6/1/2019	636	6	0
2019	7	7/1/2019	763	6	0
2019	8	8/1/2019	805	6	0
2019	9	9/1/2019	612	6	0
2019	10	10/1/2019	679	7	0
2019	11	11/1/2019	624	7	0
2019	12	12/1/2019	700	7	0
2020	1	1/1/2020	597	7	0
2020	2	2/1/2020	526	7	0
2020	3	3/1/2020	563	7	0
2020	4	4/1/2020	340	7	1
2020	5	5/1/2020	448	7	1
2020	6	6/1/2020	486	7	1
2020	7	7/1/2020	517	7	1
2020	8	8/1/2020	402	7	1
2020	9	9/1/2020	406	7	1
2020	10	10/1/2020	456	7	1

2020	11	11/1/2020	417	8	1
2020	12	12/1/2020	490	8	1
2021	1	1/1/2021	401	8	1
2021	2	2/1/2021	422	8	1
2021	3	3/1/2021	485	8	1
2021	4	4/1/2021	432	8	1
2021	5	5/1/2021	412	8	1
2021	6	6/1/2021	549	8	1
2021	7	7/1/2021	527	8	1
2021	8	8/1/2021	631	8	1
2021	9	9/1/2021	599	8	1
2021	10	10/1/2021	557	9	1
2021	11	11/1/2021	613	9	1
2021	12	12/1/2021	733	9	1
2022	1	1/1/2022	390	9	1
2022	2	2/1/2022	481	9	1
2022	3	3/1/2022	552	10	1
2022	4	4/1/2022	539	10	1
2022	5	5/1/2022	574	10	1
2022	6	6/1/2022	758	10	1
2022	7	7/1/2022	595	10	1
2022	8	8/1/2022	562	10	1
2022	9	9/1/2022	637	10	1

2022	10	10/1/2022	602	10	1
2022	11	11/1/2022	430	10	1
2022	12	12/1/2022	541	10	1
2023	1	1/1/2023	496	10	1
2023	2	2/1/2023	447	11	1
2023	3	3/1/2023	427	11	1
2023	4	4/1/2023	582	11	1
2023	5	5/1/2023	740	11	1
2023	6	6/1/2023	724	12	1
2023	7	7/1/2023	657	12	1
2023	8	8/1/2023	625	12	1
2023	9	9/1/2023	685	12	1
2023	10	10/1/2023	658	12	1
2023	11	11/1/2023	632	12	1
2023	12	12/1/2023	543	12	1
2024	1	1/1/2024	502	12	1
2024	2	2/1/2024	403	12	1
2024	3	3/1/2024	427	12	1
2024	4	4/1/2024	567	12	1
2024	5	5/1/2024	563	12	1
2024	6	6/1/2024	573	12	1
2024	7	7/1/2024	600	12	1
2024	8	8/1/2024	677	12	1

2024	9	9/1/2024	590	12	1
2024	10	10/1/2024	428	12	1
2024	11	11/1/2024	368	12	1
2024	12	12/1/2024	607	12	1
2025	1	1/1/2025	583	12	1
2025	2	2/1/2025	550	12	1
2025	3	3/1/2025	580	12	1
2025	4	4/1/2025	522	12	1
2025	5	5/1/2025	598	12	1
2025	6	6/1/2025	594	12	1
2025	7	7/1/2025	698	12	1
2025	8	8/1/2025	645	12	1
2025	9	9/1/2025	741	12	1
2025	10	10/1/2025	626	12	1
2025	11	11/1/2025	501	12	1
2025	12	12/1/2025	411	12	1
2026	1	1/1/2026	386	12	1
2026	2	2/1/2026	406	12	1
2026	3	3/1/2026	589	12	1

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.472613345
R Square	0.223363374
Adjusted R Squ	0.204872025
Standard Error	94.05312445
Observations	87

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	213707.5342	106853.7671	12.07934493	2.44991E-05
Residual	84	743063.1784	8845.99022		
Total	86	956770.7126			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
Intercept	509.7049004	45.31123617	11.24897362	1.92973E-18	419.5985381	599.8112626	419.5985381	599.8112626
HELIPT	19.92110932	5.977201644	3.332848798	0.001280023	8.034788345	31.80743029	8.034788345	31.80743029
PAND	-174.7924744	35.57146479	-4.913839659	4.35825E-06	-245.5302332	-104.0547157	-245.5302332	-104.0547157

RESIDUAL OUTPUT

Observation	*Predicted HEL*	*Residuals*	*andard Residuals*
1	629.2315563	-40.23155627	-0.432815881
2	629.2315563	-28.23155627	-0.30371845
3	629.2315563	-50.23155627	-0.540397074
4	629.2315563	-15.23155627	-0.163862899
5	629.2315563	40.76844373	0.43859178
6	629.2315563	6.768443726	0.072815725
7	629.2315563	133.7684437	1.439096873
8	629.2315563	175.7684437	1.890937883
9	629.2315563	-17.23155627	-0.185379138
10	649.1526656	29.84733441	0.321101184
11	649.1526656	-25.15266559	-0.270595377
12	649.1526656	50.84733441	0.547021689
13	649.1526656	-52.15266559	-0.561064597
14	649.1526656	-123.1526656	-1.324891066
15	649.1526656	-86.15266559	-0.926840653
16	474.3601912	-134.3601912	-1.445462963
17	474.3601912	-26.36019117	-0.283586081
18	474.3601912	11.63980883	0.125222452
19	474.3601912	42.63980883	0.458724149
20	474.3601912	-72.36019117	-0.778459568

21	474.3601912	-68.36019117	-0.735427091
22	474.3601912	-18.36019117	-0.197521127
23	494.2813005	-77.28130048	-0.831401449
24	494.2813005	-4.281300485	-0.046058741
25	494.2813005	-93.28130048	-1.003531357
26	494.2813005	-72.28130048	-0.777610852
27	494.2813005	-9.281300485	-0.099849338
28	494.2813005	-62.28130048	-0.670029659
29	494.2813005	-82.28130048	-0.885192045
30	494.2813005	54.71869952	0.588670296
31	494.2813005	32.71869952	0.351991672
32	494.2813005	136.7186995	1.470836077
33	494.2813005	104.7186995	1.12657626
34	514.2024098	42.7975902	0.46042158
35	514.2024098	98.7975902	1.06287626
36	514.2024098	218.7975902	2.353850573
37	514.2024098	-124.2024098	-1.336184339
38	514.2024098	-33.2024098	-0.357195485
39	534.1235191	17.87648088	0.192317314
40	534.1235191	4.876480883	0.052461763
41	534.1235191	39.87648088	0.428995938
42	534.1235191	223.8764809	2.408489885
43	534.1235191	60.87648088	0.654916443

44	534.1235191	27.87648088	0.299898506
45	534.1235191	102.8764809	1.106757452
46	534.1235191	67.87648088	0.730223278
47	534.1235191	-104.1235191	-1.120173238
48	534.1235191	6.876480883	0.073978002
49	534.1235191	-38.12351912	-0.410137366
50	554.0446284	-107.0446284	-1.151598881
51	554.0446284	-127.0446284	-1.366761266
52	554.0446284	27.95537157	0.300747222
53	554.0446284	185.9553716	2.000530068
54	573.9657377	150.0342623	1.614086489
55	573.9657377	83.03426225	0.893292498
56	573.9657377	51.03426225	0.549032681
57	573.9657377	111.0342623	1.194519837
58	573.9657377	84.03426225	0.904050617
59	573.9657377	58.03426225	0.624339516
60	573.9657377	-30.96573775	-0.3331331
61	573.9657377	-71.96573775	-0.774215991
62	573.9657377	-170.9657377	-1.839269799
63	573.9657377	-146.9657377	-1.581074937
64	573.9657377	-6.965737749	-0.074938238
65	573.9657377	-10.96573775	-0.117970715
66	573.9657377	-0.965737749	-0.010389522
67	573.9657377	26.03426225	0.280079699

68	573.9657377	103.0342623	1.108454883
69	573.9657377	16.03426225	0.172498506
70	573.9657377	-145.9657377	-1.570316817
71	573.9657377	-205.9657377	-2.215803974
72	573.9657377	33.03426225	0.355386534
73	573.9657377	9.034262251	0.097191671
74	573.9657377	-23.96573775	-0.257826265
75	573.9657377	6.034262251	0.064917313
76	573.9657377	-51.96573775	-0.559053605
77	573.9657377	24.03426225	0.25856346
78	573.9657377	20.03426225	0.215530983
79	573.9657377	124.0342623	1.334375388
80	573.9657377	71.03426225	0.764195066
81	573.9657377	167.0342623	1.796974517
82	573.9657377	52.03426225	0.5597908
83	573.9657377	-72.96573775	-0.78497411
84	573.9657377	-162.9657377	-1.753204845
85	573.9657377	-187.9657377	-2.022157827
86	573.9657377	-167.9657377	-1.806995442
87	573.9657377	15.03426225	0.161740387

ANO	MES	DATA	HEL	HELIPT	QAVREF
2022	3	3/1/2022	552	10	4.968086
2022	4	4/1/2022	539	10	4.915976
2023	7	7/1/2023	657	12	3.927309
2023	8	8/1/2023	625	12	4.072688
2023	9	9/1/2023	685	12	4.706179
2023	10	10/1/2023	658	12	5.091458
2023	11	11/1/2023	632	12	5.031612
2023	12	12/1/2023	543	12	4.763164
2024	1	1/1/2024	502	12	4.248213
2024	2	2/1/2024	403	12	4.188744
2024	3	3/1/2024	427	12	4.457176
2024	4	4/1/2024	567	12	4.450559
2024	5	5/1/2024	563	12	4.513989
2024	6	6/1/2024	573	12	4.266493
2024	7	7/1/2024	600	12	4.315163
2024	8	8/1/2024	677	12	4.495767
2024	9	9/1/2024	590	12	4.335217
2024	10	10/1/2024	428	12	3.984499
2024	11	11/1/2024	368	12	3.91231
2024	12	12/1/2024	607	12	4.00599
2025	1	1/1/2025	583	12	4.18173
2025	2	2/1/2025	550	12	4.528818

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.17794485
R Square	0.03166437
Adjusted R Squ	0.002320866
Standard Error	95.74562231
Observations	35

ANOVA

	df
Regression	1
Residual	33
Total	34

	Coefficients
Intercept	405.5300002
QAVREF	37.2289583

RESIDUAL OUTPUT

2025	3	3/1/2025	580	12	4.249874
2025	4	4/1/2025	522	12	3.89073
2025	5	5/1/2025	598	12	3.859696
2025	6	6/1/2025	594	12	3.621838
2025	7	7/1/2025	698	12	3.638104
2025	8	8/1/2025	645	12	3.790171
2025	9	9/1/2025	741	12	3.667032
2025	10	10/1/2025	626	12	3.670938
2025	11	11/1/2025	501	12	3.703496
2025	12	12/1/2025	411	12	3.833291
2026	1	1/1/2026	386	12	3.532179
2026	2	2/1/2026	406	12	3.442872
2026	3	3/1/2026	589	12	3.65865

Observation	*Predicted HEL*
1	590.486685
2	588.5466751
3	551.7396395
4	557.1519356
5	580.7361418
6	595.0796909
7	592.8516643
8	582.8576365
9	563.6865398
10	561.4725597
11	571.4660356
12	571.2196652
13	573.5811211
14	564.3670881
15	566.1790115
16	572.9027157
17	566.9256067
18	553.8687308
19	551.1812078
20	554.6688525
21	561.2114424
22	574.1331646
23	563.7483701
24	550.3778377
25	549.2224625

26	540.3672674
27	540.9728181
28	546.6341272
29	542.0497838
30	542.195195
31	543.4072862
32	548.2394298
33	537.0293328
34	533.7045507
35	541.7377285

SS	*MS*	*F*	*Significance F*
9892.287405	9892.287405	1.079092995	0.306452343
302518.3983	9167.224191		
312410.6857			

Standard Error	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
150.2903191	2.69831086	0.010897684	99.7620469	711.2979536	99.7620469	711.2979536
35.83863398	1.03879401	0.306452343	-35.68529078	110.1432074	-35.68529078	110.1432074

Residuals	*andard Residuals*
-38.48668498	-0.408013063
-49.54667514	-0.525264535
105.2603605	1.115908063
67.8480644	0.719285035
104.2638582	1.105343735
62.92030908	0.667043889
39.14833567	0.415027492
-39.85763651	-0.42254708
-61.68653979	-0.653964197
-158.4725597	-1.680032314
-144.4660356	-1.531543433
-4.219665224	-0.044734394
-10.58112107	-0.112174785
8.632911919	0.091521024
33.82098848	0.358550109
104.0972843	1.103577817
23.07439326	0.244621065
-125.8687308	-1.334385811
-183.1812078	-1.941978782
52.33114752	0.554783863
21.78855758	0.230989396
-24.13316457	-0.255845532
16.25162992	0.172290165
-28.37783769	-0.300845045
48.77753751	0.517110592

53.63273259	0.568582498
157.0271819	1.664709275
98.36587277	1.042816783
198.9502162	2.109152479
83.80480501	0.888448958
-42.40728622	-0.449576957
-137.2394298	-1.454931235
-151.0293328	-1.601123626
-127.7045507	-1.353848087
47.2622715	0.501046639